0 - 26 40% → 50% ⇒ 45-60%

7-8 10

DAC ⇓ recur

Lec_27

Recurrence Relation Containing Root

DAC
⇓
Recursion
⇓
RR solve
⇓
S.T.M.
R.T.M.
M.T.

$T(n) = T(\sqrt{n}) + n^2$

Question1: T(n)=T(√n)+c ,find TC?

$T(n) = T(\sqrt{n}) + C$ How to solve?

Base? : $n=1,\ T(1)=0$
$n=0,\ T(0)=1$

Solⁿ

$T(n) = T(\sqrt{n}) + C$ → let

$\boxed{\text{let } n = 2^x}$

$\Rightarrow T(2^x) = T((2^x)^{1/2}) + C$

$\Rightarrow T(2^x) = T(2^{x/2}) + C$

$\boxed{\text{let } T(2^x) = S(x)}$

$T(2^x) = S(x)$

$T(2^{x/2}) = S(x/2)$

putting $x = \frac{x}{2}$

$\Rightarrow \boxed{S(x) = S\left(\frac{x}{2}\right) + C}$

Relate

$T(n) = T\left(\frac{n}{2}\right) + C$

apply master theorem

$f(n) = \cdot\ ,\quad a=1,\quad b=2$

$T(n) = T(\sqrt{n}) + c$

$f(n)$	$n^{\log_b a}$
c	$n^{\log_2 1}$
c	n^0
c	1

$TC =$ case 3 of masters

$TC = \theta(c \log n)$

$TC = \theta(\log n)$

$TC = \theta(\log \log n)$

$TC = \theta(\log \log n)$

we know

$n = 2^x$

$\boxed{\log n = x}$

Method 2)

$n = 2^x$

$T(2^x) = T(2^{x/2}) + c$

let $T(2^x) = S(x)$

$S(x) = S(x/2) + c$

Question2: T(n)=2T(√n)+logn ,find TC?

$T(n) = 2T(\sqrt{n}) + \log_2 n$

Let $n = 2^x$

$T(2^x) = 2T(2^{x/2}) + \log_2 2^x$

$T(2^x) = 2T(2^{x/2}) + x$

Let $T(2^x) = S(x)$

$T(2^x) = S(x)$

$x = x/2$

$T(2^{x/2}) = S(\frac{x}{2})$

$S(x) = 2S(x/2) + x$

Solve By master theorem.

$a = 2 \quad b = 2$

$f(x) = x$

$f(x)$	$x^{\log_b a}$
x	$x^{\log_2 2}$

TC =
$O(\log n \times \log\log n)$

$n \mid n$. case: 3

$x \log x$

we know that, $n = 2^x$

$x = \log n$

$\log n \cdot \log\log n$

Master theorem

Practice makes you perfect

⇒ SD - GOI

S-G

THANKS FOR WATCHING

- **WHAT NEXT?**

DAC ⇒ [illegible]